BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The brain

The written notes for the brain lecture: the four divisions, brain development, the cerebral surface and lobes, functional cortical areas, white matter tracts, basal nuclei, diencephalon, cerebellum, limbic system, and the arterial supply through the Circle of Willis. Module 5 competencies 151 to 160, with ten worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 151. Brain gross organization

In your course notes: Module 5 packet, Chapter 4, competency 151 (page 29).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Identify the four major brain divisions cerebrum, diencephalon, brainstem, and cerebellum, and distinguish the cortical gray matter from the deep white matter on a specimen.

The brain sits in the cranial cavity. At the foramen magnum it continues downward as the spinal cord. Anatomists divide the brain into four major divisions, and you can find all four on a midsagittal section of the brain model (a cut straight down the midline, viewed from the side).

The four divisions

Division	Where it is	Main parts	Main job
Cerebrum	Upper cranial cavity; the largest division	Two cerebral hemispheres, each with cortex, white matter, and basal nuclei	Conscious thought, sensation, and voluntary movement
Diencephalon	Deep at the core, between the hemispheres and the brainstem	Thalamus, hypothalamus, epithalamus	Relays sensory signals to the cortex and controls homeostasis
Brainstem	Below the diencephalon, in front of the cerebellum; continuous with the spinal cord	Midbrain, pons, medulla oblongata	Connects the brain to the spinal cord
Cerebellum	Behind the brainstem, below the occipital lobes, in the posterior cranial fossa	Two cerebellar hemispheres joined by the vermis	Coordinates movement

- The cerebrum is the largest division. It forms the two cerebral hemispheres that fill the upper cranial cavity.
- The diencephalon sits at the core of the brain, between the hemispheres and above the brainstem. The hemispheres cover it, so you see it best on a midsagittal cut.

- The brainstem is the stalk below the diencephalon. It is made of the midbrain, pons, and medulla oblongata, and it continues as the spinal cord.
- The cerebellum sits behind the brainstem and below the occipital lobes, in the posterior cranial fossa. It is the second largest division, bigger than the brainstem and the diencephalon.

Gray matter and white matter

The brain is built from two kinds of nervous tissue. Gray matter holds neuron cell bodies, dendrites, unmyelinated axons, and synapses. It looks gray to tan. White matter is made of myelinated axons running in bundles called tracts. Myelin is fatty, so white matter looks pale.

Cerebral cortex

In the cerebrum, gray matter forms a thin outer layer called the cerebral cortex. It is only about 2 to 4 mm thick, and it follows every gyrus and sulcus.

Cerebral white matter

Beneath the cortex lies the deep core of tracts. Islands of gray matter also sit deep inside the white matter. These are the basal nuclei and the thalamus.

The same plan in the cerebellum

Gray cortex on the outside and white matter inside.

The opposite plan in the brainstem and spinal cord

Most of their white matter is on the outside, with gray matter inside.

- On a specimen, look at a coronal or horizontal slice. The cortex is the thin, darker rim that traces the outline of each fold. The pale tissue deep to it is white matter. Darker patches buried inside the white matter are the deep nuclei.
- Gray matter is the thin outer layer of the cerebrum and white matter is the core. Students reverse these on drawings often, so check yours.
- Memory tip: Cortex is Latin for bark. Like bark on a tree, the cortex is a thin outer layer, and the white matter is the wood inside.

Competency 152. Neural tube brain development

In your course notes: Module 5 packet, Chapter 4, competency 152 (page 30).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Trace brain development from the neural tube through the primary and secondary vesicles and match each secondary vesicle to its adult structures and ventricle.

The whole central nervous system develops from the neural tube, which forms from ectoderm. In the early embryo a strip of ectoderm called the neural plate folds into a groove, and the edges of the groove fuse into a tube. The front (cranial) end of the tube becomes the brain. The rest of the tube becomes the spinal cord.

The tube stays hollow. Its central cavity becomes the ventricles of the brain and the central canal of the spinal cord. So as you follow each part of the tube into the adult brain, you also follow its cavity into a ventricle.

From tube to adult brain, in order

1. Neural tube: forms from ectoderm. The cranial end becomes the brain and the rest becomes the spinal cord.
2. Three primary vesicles, by about the fourth week: the cranial end of the tube swells into the prosencephalon (forebrain), the mesencephalon (midbrain), and the rhombencephalon (hindbrain).
3. Five secondary vesicles, by about the fifth week: the prosencephalon divides into the telencephalon and the diencephalon. The mesencephalon does not divide. The rhombencephalon divides into the metencephalon and the myelencephalon.
4. Adult brain: each secondary vesicle becomes a set of adult structures and keeps part of the tube cavity as a ventricle.

Secondary vesicle	Primary vesicle it came from	Adult structures	Cavity in the adult
Telencephalon	Prosencephalon	Cerebrum (cerebral hemispheres, with cortex, white matter, and basal nuclei)	Lateral ventricles
Diencephalon	Prosencephalon	Thalamus, hypothalamus, epithalamus	Third ventricle
Mesencephalon	Mesencephalon (unchanged)	Midbrain	Cerebral aqueduct
Metencephalon	Rhombencephalon	Pons and cerebellum	Upper part of the fourth ventricle
Myelencephalon	Rhombencephalon	Medulla oblongata	Lower part of the fourth ventricle, which continues into the central canal of the spinal cord

- There are five secondary vesicles because the mesencephalon carries through without splitting. If your chart has only four, you dropped it.
- The cerebellum grows from the metencephalon along with the pons, not from the myelencephalon.
- The telencephalon grows far more than the other vesicles. The cerebral hemispheres expand backward over the diencephalon and midbrain. That growth is why the diencephalon ends up hidden deep in the adult brain.
- Memory tip: Read the five secondary vesicles in order from the front of the tube to the back and you name the adult brain from top to bottom: cerebrum, diencephalon, midbrain, pons and cerebellum, medulla. The prefixes help too: tel means end, di means between, mes means middle, met means after, and myel means marrow, for the part next to the spinal cord.

Competency 153. Cerebral surface and lobes

In your course notes: Module 5 packet, Chapter 4, competency 153 (page 31).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Identify the cerebral hemispheres, five lobes, gyri, sulci, and the named central, lateral, and parieto-occipital sulci along with the longitudinal and transverse fissures on a specimen.

The cerebrum is made of two cerebral hemispheres, right and left. A deep midline cleft called the longitudinal fissure separates them. At the bottom of that fissure, the two hemispheres are joined by the corpus callosum. A second deep cleft, the transverse fissure, separates the back of the cerebrum from the cerebellum below it.

Gyri (singular, gyrus)
: The raised ridges of the folded surface.

Sulci (singular, sulcus)
: The grooves between the gyri.

Fissures
: The deepest grooves. Folding packs much more cortex into the skull than a smooth surface could hold.

The named grooves

A few sulci are named because they mark the lobe boundaries. The central sulcus runs from the top of the hemisphere down the side toward the lateral sulcus. The precentral gyrus lies just in front of it, bounded in front by the precentral sulcus. The postcentral gyrus lies just behind it, bounded behind by the postcentral sulcus. The lateral sulcus (Sylvian fissure) is the deep groove on the side of each hemisphere. The parieto-occipital sulcus is easiest to see on the medial surface, near the back of the hemisphere.

Groove	Where to find it	What it separates
Longitudinal fissure	Midline, between the hemispheres	The right and left cerebral hemispheres
Transverse fissure	Back of the brain, between the occipital lobes and the cerebellum	The cerebrum from the cerebellum
Central sulcus	Top and side of each hemisphere, running down toward the lateral sulcus	The frontal lobe from the parietal lobe; the precentral gyrus in front from the postcentral gyrus behind
Precentral sulcus	Just in front of the precentral gyrus	The precentral gyrus from the rest of the frontal lobe
Postcentral sulcus	Just behind the postcentral gyrus	The postcentral gyrus from the rest of the parietal lobe
Lateral sulcus (Sylvian fissure)	Deep groove on the side of each hemisphere	The temporal lobe below from the frontal and parietal lobes above
Parieto-occipital sulcus	Medial surface, near the back of the hemisphere	The parietal lobe from the occipital lobe

The five lobes

Each hemisphere has five lobes, so every lobe is paired, with one in each hemisphere. Four lobes are named for the cranial bone that covers them: the frontal, parietal, temporal, and occipital bones. The fifth, the insula, is hidden deep in the lateral sulcus, so no outside view shows it. To see it, gently pull the temporal lobe away from the frontal and parietal lobes and look into the lateral sulcus.

Lobe	Location	Boundaries
Frontal lobe	Front of each hemisphere, under the frontal bone	Central sulcus behind; lateral sulcus below
Parietal lobe	Top and upper side, under the parietal bone	Central sulcus in front; parieto-occipital sulcus behind; lateral sulcus below
Temporal lobe	Lower side, under the temporal bone	Lateral sulcus above
Occipital lobe	Back of each hemisphere, under the occipital bone	Parieto-occipital sulcus in front; transverse fissure below, separating it from the cerebellum
Insula	Deep inside the lateral sulcus	Covered by the frontal, parietal, and temporal lobes

- On a specimen, find the lateral sulcus first, because it is the deepest groove on the side. Then look for the central sulcus. It usually runs without a break from the top of the hemisphere almost down to the lateral sulcus, with a long, unbroken gyrus on each side of it. Those two gyri are the precentral and postcentral gyri.
- Seen from above, the longitudinal fissure runs down the middle and the central sulcus crosses each hemisphere. You can see the frontal, parietal, and occipital lobes from the top; the temporal lobe is on the side and the insula is hidden.
- Memory tip: Pre means before and post means after. The precentral gyrus comes before the central sulcus (in front of it) and the postcentral gyrus comes after it (behind it). These two get swapped a lot.

Competency 154. Functional cortical areas

In your course notes: Module 5 packet, Chapter 4, competency 154 (page 33).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Map the motor, sensory, and association areas of the cortex to their gyri and lobes, including primary motor, primary somatosensory, primary

visual, auditory, gustatory, and olfactory cortices, Broca and Wernicke areas, and the prefrontal cortex.

The cortex is mapped into areas that each have a fixed location. There are three kinds. Motor areas control voluntary movement and sit in the frontal lobe. Sensory areas are the first cortex to receive a given kind of sensation. A primary sensory area receives that input, and an association area beside it interprets it. Association areas make up the rest of the cortex. They interpret input, connect it with memory, and plan responses.

Motor areas, frontal lobe

- The primary motor cortex is the precentral gyrus. Its neurons send commands for voluntary movement to skeletal muscles on the opposite side of the body.
- The body is mapped along the gyrus upside down. The leg and foot are on the medial surface, inside the longitudinal fissure. The hand is higher on the lateral surface, and the face is lowest, near the lateral sulcus. Body parts with fine control, such as the hand, lips, and tongue, get the largest share of cortex.
- The premotor cortex lies just in front of the precentral gyrus and plans learned, patterned movements.
- Broca's area sits in the inferior frontal gyrus, usually in the left hemisphere. It plans the movements of speech, so it counts as a motor area.

Sensory areas

- The primary somatosensory cortex (primary sensory cortex) is the postcentral gyrus of the parietal lobe. It receives touch, pressure, pain, temperature, and body position from the opposite side of the body. Its body map is laid out in the same order as the motor map next to it. The somatosensory association area lies just behind it.
- The primary visual cortex is in the occipital lobe, along the calcarine sulcus on the medial surface.
- The primary auditory cortex is on the superior temporal gyrus, along the lower edge of the lateral sulcus.
- The primary gustatory cortex (taste) is in the insula and the lower end of the postcentral gyrus that covers it.
- The primary olfactory cortex (smell) is on the medial surface of the temporal lobe, near the uncus. Smell reaches it from the olfactory bulb and olfactory tract, which lie on the inferior surface of the frontal lobe.

Association areas

- Wernicke's area sits in the posterior part of the superior temporal gyrus, where it meets the parietal lobe, usually in the left hemisphere. It is needed to understand spoken and written language. A tract called the arcuate fasciculus connects it to Broca's area.
- The prefrontal cortex covers the front of the frontal lobe. It is a large association area that handles planning, judgment, and personality.

Area	Kind	Lobe	Gyrus or landmark	Main job
Primary motor cortex	Motor	Frontal	Precentral gyrus	Sends commands for voluntary movement
Premotor cortex	Motor	Frontal	Just in front of the precentral gyrus	Plans learned, patterned movements
Frontal eye field	Motor	Frontal	Middle frontal gyrus	Controls voluntary eye movement
Broca's area	Motor	Frontal, usually left	Inferior frontal gyrus	Plans the movements of speech
Primary somatosensory cortex	Primary sensory	Parietal	Postcentral gyrus	Receives touch, pressure, pain, temperature, and body position
Somatosensory association area	Association	Parietal	Just behind the postcentral gyrus	Interprets texture, shape, and position
Primary visual cortex	Primary sensory	Occipital	Calcarine sulcus, medial surface	Receives visual input
Visual association area	Association	Occipital	Surrounding the primary visual cortex	Recognizes objects and faces
Primary auditory cortex	Primary sensory	Temporal	Superior temporal gyrus, at the lateral sulcus	Receives sound input
Auditory association area	Association	Temporal	Next to the primary auditory cortex	Interprets speech and music
Primary gustatory cortex	Primary sensory	Insula and parietal	Insula and the lower end of the postcentral gyrus	Receives taste input
Primary olfactory cortex	Primary sensory	Temporal	Medial temporal lobe, near the uncus	Receives smell input
Wernicke's area	Association	Temporal, usually left	Posterior superior temporal gyrus, next to the parietal lobe	Understanding spoken and written language
Prefrontal cortex	Association	Frontal	Front of the frontal lobe	Planning, judgment, and personality

- The thalamus relays most sensory input up to these primary areas. It is a relay station deep in the diencephalon, below the cortex, and competency 157 covers it.
- Broca's area is in the frontal lobe and Wernicke's area is back where the temporal and parietal lobes meet. Speaking is in front, understanding is behind.
- Memory tip: Front moves, back feels. The precentral gyrus, in front of the central sulcus, moves the body. The postcentral gyrus, behind it, feels the body.

Competency 155. Cerebral white matter tracts

In your course notes: Module 5 packet, Chapter 4, competency 155 (page 34).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Compare association, commissural, and projection fibers by what each connects and identify the corpus callosum and internal capsule on a specimen.

The cerebral white matter is made of myelinated axons grouped into tracts. The tracts fall into three classes, sorted by what they connect.

Association fibers

Stay within one hemisphere. Short ones arc from one gyrus to the next. Long ones connect one lobe to another. The arcuate fasciculus, which links Broca's area and Wernicke's area, is an example. Because it stays in one hemisphere, it is an association tract, not a commissural one.

Commissural fibers

Cross the midline and connect matching areas of the right and left hemispheres. The largest is the corpus callosum, a thick, arched band of white matter at the bottom of the longitudinal fissure. On a midsagittal section it is the C-shaped white band above the diencephalon, and it forms the roof of the lateral ventricles. The anterior commissure and posterior commissure are smaller commissural bundles.

Projection fibers

Connect the cortex with the lower brain and spinal cord. Some carry signals down from the cortex, and some carry signals up to it. Just below the cortex they spread out in a fan called the corona radiata. Deeper down they gather into one compact band, the internal capsule. Below that, the descending fibers continue into the midbrain as the crus cerebri.

- Hanging below the corpus callosum is the septum pellucidum, a thin sheet that separates the two lateral ventricles. The fornix runs along its lower edge.
- The internal capsule passes between the thalamus and caudate nucleus on its medial side and the lentiform nucleus (putamen and globus pallidus) on its lateral side.

Fiber class	What it connects	Examples	Where to see it
Association fibers	Gyri and lobes within the same hemisphere	Arcuate fasciculus (Broca area to Wernicke area)	Inside one hemisphere; they never cross the midline
Commissural fibers	Matching areas of the right and left hemispheres	Corpus callosum (largest); anterior and posterior commissures	Corpus callosum on a midsagittal section, the arched band above the diencephalon
Projection fibers	The cortex with the diencephalon, brainstem, and spinal cord, in both directions	Corona radiata; internal capsule; crus cerebri below it	Internal capsule on a coronal or horizontal section, between the thalamus and caudate nucleus and the lentiform nucleus

- On a specimen, find the corpus callosum on the midsagittal cut first. To find the internal capsule, use a horizontal or coronal section. On a horizontal section it is a white band bent into a V that points toward the midline, with the caudate nucleus and thalamus on its inner side and the lentiform nucleus on its outer side.
- Direction is the quickest check. Association fibers stay on one side, commissural fibers cross side to side between the hemispheres, and projection fibers run up and down.
- Memory tip: Commissure means a joining, so commissural fibers join the two hemispheres. Projection fibers project down to the brainstem and spinal cord or up from them. Association fibers stay in one hemisphere.

What this means for patients

- The projection fibers of the internal capsule carry nearly all the traffic between the cortex and the body through one narrow band. A small stroke there can cut a great deal of wiring at once, which is why such a small lesion can weaken the entire opposite side of the body.

Competency 156. Basal ganglia

In your course notes: Module 5 packet, Chapter 4, competency 156 (page 35).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Identify the caudate nucleus, putamen, and globus pallidus and locate the functionally grouped substantia nigra and subthalamic nucleus.

The basal nuclei, also called the basal ganglia, are paired masses of gray matter buried deep in the cerebral white matter of each hemisphere. They lie beside the lateral ventricle and the thalamus. Clusters of cell bodies in the central nervous system are usually called nuclei, so many texts use basal nuclei. Basal ganglia is the older name and is still widely used. The basal nuclei work with the motor cortex. They share three roles: they help start and

regulate wanted movements, they hold back unwanted movements, and they set muscle tone.

The three core nuclei, in the cerebrum

Caudate nucleus
: Long and curved, with a head, a body, and a tail. Its large head bulges into the lateral wall of the front part (anterior horn) of the lateral ventricle. The body and tail follow the curve of the ventricle backward and then down into the temporal lobe.

Putamen
: The outer, lateral part of the lentiform nucleus.

Globus pallidus
: The inner, medial part of the lentiform nucleus, between the putamen and the internal capsule. It looks paler.

Lentiform nucleus
: The putamen and globus pallidus together, a lens-shaped mass lateral to the internal capsule.

Striatum
: The caudate nucleus and putamen together, named because bands of gray matter cross the internal capsule between them and give the area a striped look.

Two nuclei grouped with them by function

Two more nuclei sit outside the cerebrum but work in the same movement circuits, so they are grouped with the basal nuclei by function. The substantia nigra is a band of dark, pigmented neurons in the midbrain, just behind the crus cerebri. Loss of these neurons is the change seen in Parkinson disease. The subthalamic nucleus is a small nucleus in the diencephalon, below the thalamus and just above the substantia nigra.

Structure	Brain region	Where it is	How to find it on a section
Caudate nucleus	Cerebrum	Curves along the lateral ventricle; head in the wall of the anterior horn	Gray mass bulging into the lateral ventricle, medial to the internal capsule
Putamen	Cerebrum	Lateral part of the lentiform nucleus	The outer, darker part of the lens-shaped mass lateral to the internal capsule
Globus pallidus	Cerebrum	Medial part of the lentiform nucleus	The inner, paler part of that mass, between the putamen and the internal capsule
Substantia nigra	Midbrain	Just behind the crus cerebri	Dark band across the midbrain on a cross section
Subthalamic nucleus	Diencephalon	Below the thalamus, above the substantia nigra	Small nucleus under the thalamus on a coronal section

- Sorting by region: the caudate nucleus, putamen, and globus pallidus are in the cerebrum; the subthalamic nucleus is in the diencephalon; the substantia nigra is in the midbrain. The last two work with the basal nuclei but do not sit in the cerebrum.
- The globus pallidus is medial, closer to the midline, and the putamen is lateral. Check your section.
- Memory tip: Pallidus means pale. The globus pallidus is the paler, inner part of the lentiform nucleus, and the putamen wraps around it on the outside.

What this means for patients

- The substantia nigra is darkly pigmented, and the loss of its neurons is the change seen in Parkinson disease. Because it works in the same movement circuits as the basal nuclei, losing it disturbs the starting, holding back, and tone-setting of movement that those circuits handle.

Competency 157. Diencephalon

In your course notes: Module 5 packet, Chapter 4, competency 157 (page 36).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Identify the thalamus with the massa intermedia, the hypothalamus, the epithalamus with the pineal gland, and the third ventricle they enclose.

The diencephalon lies between the cerebrum and the midbrain. The cerebral hemispheres cover it, so you see it best on a midsagittal section. It has three parts, the thalamus, the hypothalamus, and the epithalamus, and together they surround the third ventricle.

Thalamus

- A pair of egg-shaped masses of gray matter, one on each side of the third ventricle. It makes up about 80 percent of the diencephalon.
- It is made of many nuclei, and it relays most sensory input on its way to the cerebral cortex.
- The massa intermedia (interthalamic adhesion) is a small bridge of gray matter that crosses the third ventricle and joins the two thalami. On a midsagittal section it shows up as a small cut oval in the middle of the thalamus. Some people do not have one. It is part of the thalamus, not the hypothalamus.

Hypothalamus

- Lies below the thalamus. A shallow groove on the wall of the third ventricle, the hypothalamic sulcus, marks the border between them. The hypothalamus forms the floor and the lower walls of the third ventricle.
- It is the main control center for homeostasis. It controls the pituitary gland and the autonomic nervous system and helps regulate body temperature, hunger, thirst, and sleep and wake cycles.
- On the inferior surface of the brain, three landmarks belong to this area. The optic chiasm sits at its front, where the optic nerves meet and some of their fibers cross before continuing as the optic tracts. The infundibulum is the stalk behind the chiasm that connects the hypothalamus to the pituitary gland. The mammillary bodies are two small round bumps behind the infundibulum.

Epithalamus

- The small, posterior, upper part of the diencephalon. It forms part of the roof of the third ventricle.
- Its main structure is the pineal gland, a small pine-cone-shaped endocrine gland that projects backward from the roof of the third ventricle, above the midbrain. The pineal gland secretes melatonin.
- The epithalamus also holds the habenular nuclei.

Third ventricle

The third ventricle is the narrow, midline cavity enclosed by these parts. The thalami form its upper side walls, the hypothalamus forms its floor and lower walls, and the epithalamus forms the back of its roof. In front, it connects to each lateral ventricle through an interventricular foramen. Behind, it drains into the cerebral aqueduct of the midbrain.

Part	Position	Features you identify	Relation to the third ventricle
Thalamus	Upper and largest part; one on each side	Paired thalami; massa intermedia joining them	Forms the upper side walls
Hypothalamus	Below the thalamus	Optic chiasm in front, infundibulum, mammillary bodies	Forms the floor and lower side walls
Epithalamus	Back and top	Pineal gland; habenular nuclei	Forms the back of the roof
Third ventricle	Midline, between the two halves of the diencephalon	Narrow slit crossed by the massa intermedia	The cavity itself; joins the lateral ventricles in front and the cerebral aqueduct behind

- On a midsagittal view, the corpus callosum arches above the diencephalon, the optic chiasm and infundibulum hang below the hypothalamus at the front, the pineal gland projects back from the epithalamus, and the midbrain sits below.
- Memory tip: Hypo means below, so the hypothalamus sits below the thalamus. Epi means upon, so the epithalamus sits on top at the back.

Competency 158. Cerebellum

In your course notes: Module 5 packet, Chapter 4, competency 158 (page 37).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Identify the cerebellar hemispheres, vermis, folia, arbor vitae, deep nuclei, and the three cerebellar peduncles with their brainstem connections.

The cerebellum is the second largest part of the brain. It sits in the posterior cranial fossa, below the occipital lobes and behind the pons and medulla. The transverse fissure separates it from the cerebrum above. The fourth ventricle lies between the cerebellum and the brainstem. The cerebellum coordinates movement, posture, and balance, adjusting the timing and accuracy of movements without our awareness.

The outside

Cerebellar hemispheres

The two side halves, each with gray cortex over white matter.

Vermis

The narrow midline strip that joins the two hemispheres.

Folia

Thin, parallel, leaf-like folds of the surface, much finer than the gyri of the cerebrum. They are gray matter.

Lobes

A deep groove called the primary fissure divides the cerebellum into an anterior lobe and a posterior lobe, and a small flocculonodular lobe sits on its underside.

The inside

Cerebellar cortex

The outer layer of gray matter. The cerebellum follows the same plan as the cerebrum: gray outside, white inside.

Arbor vitae

The white matter under the cortex branches out into every folium. On a midsagittal cut through the vermis it looks like a branching tree, so it is called the arbor vitae, Latin for tree of life.

Deep cerebellar nuclei

Four pairs of gray matter nuclei buried in the white matter. From lateral to medial they are the dentate nucleus (the largest), the emboliform nucleus, the globose nucleus, and the fastigial nucleus. Most signals leaving the cerebellum start in these nuclei. On a shaded drawing they are gray, even though they sit inside the white arbor vitae.

The three pairs of cerebellar peduncles

Three pairs of cerebellar peduncles, thick stalks of white matter, attach the cerebellum to the brainstem. Every fiber entering or leaving the cerebellum passes through one of them. Each peduncle joins a different part of the brainstem.

Peduncle	Brainstem connection	Main traffic	Mostly in or out
Superior cerebellar peduncle	Midbrain	Mostly output, from the deep cerebellar nuclei toward the midbrain and thalamus	Out of the cerebellum
Middle cerebellar peduncle	Pons	Input from the pontine nuclei, which relay signals from the cerebral cortex; the largest peduncle	Into the cerebellum
Inferior cerebellar peduncle	Medulla oblongata	Input from the spinal cord about body position, from the vestibular nuclei about balance, and from the olive of the medulla	Into the cerebellum

- On a midsagittal specimen, find the vermis cut down the middle, the arbor vitae branching through it, and the folia along its edge. The deep nuclei and the peduncles are easiest to see on models or on sections that cut through the hemispheres.
- The superior peduncle mostly carries signals out. The middle and inferior peduncles carry signals in.
- Memory tip: The peduncles attach in order from top to bottom: superior to the midbrain, middle to the pons, inferior to the medulla.

Competency 159. Limbic system

In your course notes: Module 5 packet, Chapter 4, competency 159 (page 38).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Locate the cingulate gyrus, amygdala, and hippocampus as the limbic structures ringing the diencephalon.

The limbic system is a group of structures on the medial side of each hemisphere. Limbic comes from limbus, meaning border or ring. These structures form a ring around the diencephalon and the top of the brainstem. They work together in two jobs, emotion and memory.

The ring, structure by structure

Cingulate gyrus

The gyrus that arches just above the corpus callosum on the medial surface. Below it, the sulcus of the corpus callosum separates it from the corpus callosum. Above it, the cingulate sulcus separates it from the rest of the frontal and parietal lobes. It helps connect emotion with behavior. On a midsagittal section it is the first gyrus above the corpus callosum, so it is the part of the ring that sits above the diencephalon.

Parahippocampal gyrus

The cingulate gyrus curves around the back of the corpus callosum and continues down into the temporal lobe as the parahippocampal gyrus, on the inferior and medial surface of the temporal lobe. The front end of the parahippocampal gyrus hooks back on itself to form the uncus. Together these gyri make the C-shaped ring that gives the system its name.

Hippocampus

A curved ridge of cortex rolled into the floor of the inferior horn of the lateral ventricle (also called the temporal horn), deep to the parahippocampal gyrus in the medial temporal lobe. Its name means seahorse, which describes its shape. It is needed to form new long-term memories.

Fornix

The output of the hippocampus travels in the fornix, an arching band of white matter that runs forward under the corpus callosum, along the lower edge of the septum pellucidum, and ends at the mammillary bodies of the hypothalamus.

Amygdala

An almond-shaped nucleus deep in the front of the medial temporal lobe, just in front of the hippocampus and deep to the uncus. It is involved in fear and other emotional responses, and it links emotion to memory.

Structure	Where it is	Landmark to find it	Main role
Cingulate gyrus	Medial surface of the hemisphere, arching over the corpus callosum	First gyrus above the corpus callosum on a midsagittal section, between the sulcus of the corpus callosum and the cingulate sulcus	Connects emotion with behavior
Hippocampus	Medial temporal lobe, in the floor of the inferior horn of the lateral ventricle	Deep to the parahippocampal gyrus; its output leaves in the fornix	Forms new long-term memories
Amygdala	Front of the medial temporal lobe	Deep to the uncus, just in front of the hippocampus	Fear and emotional responses; links emotion to memory
Parahippocampal gyrus	Inferior and medial surface of the temporal lobe	Continues from the cingulate gyrus; its front end forms the uncus	Completes the limbic ring
Fornix	Arch of white matter under the corpus callosum	Lower edge of the septum pellucidum on a midsagittal section	Carries hippocampal output to the mammillary bodies

- Picture the ring on a midsagittal view: the cingulate gyrus arches over the corpus callosum above the diencephalon, and the hippocampus and amygdala curve down and forward below the diencephalon into the temporal lobe, with the amygdala at the front end.
- The hippocampus forms new memories and the amygdala handles fear. Students swap these roles often, so check yours.

- Memory tip: Amygdala means almond and hippocampus means seahorse. The names describe their shapes, and both sit in the medial temporal lobe, with the amygdala in front.

Competency 160. Cerebral arterial supply

In your course notes: Module 5 packet, Chapter 4, competency 160 (page 39).

Dr. Rennie's video for this part: The brain concept video.

You should be able to: Trace the anterior and posterior circulations through the Circle of Willis and identify the anterior, middle, and posterior cerebral arteries, the communicating arteries, the internal carotid, vertebral, and basilar arteries, and the cerebellar branches.

The brain is supplied by two pairs of arteries. The two internal carotid arteries form the anterior circulation, which feeds most of the cerebrum. The two vertebral arteries form the posterior circulation, which feeds the brainstem, the cerebellum, and the occipital lobe. The two systems join in a ring of arteries on the inferior surface of the brain, the cerebral arterial circle, also called the Circle of Willis. The ring surrounds the optic chiasm, the infundibulum, and the mammillary bodies.

Anterior circulation

- Each internal carotid artery rises through the neck, enters the skull through the carotid canal of the temporal bone, and reaches the base of the brain beside the optic chiasm. There it divides into two branches.
- The anterior cerebral artery runs forward and medially into the longitudinal fissure and arches over the corpus callosum. It supplies the medial surface of the frontal and parietal lobes, including the leg and foot areas of the motor and somatosensory cortex.
- The anterior communicating artery is a short bridge between the right and left anterior cerebral arteries, in front of the optic chiasm.
- The middle cerebral artery is the largest branch of the internal carotid. It runs laterally into the lateral sulcus and supplies most of the lateral surface of the hemisphere. That territory includes the face and upper limb areas of the motor and somatosensory cortex, the auditory cortex, the insula, and Broca's and Wernicke's areas.

Posterior circulation

- The vertebral arteries branch from the subclavian arteries and rise through the transverse foramina of the cervical vertebrae. They enter the skull through the foramen magnum and run up the front of the medulla.
- At the junction of the medulla and pons, the two vertebral arteries unite to form the basilar artery. The basilar artery runs up the front of the pons in the midline, giving off small pontine arteries to the pons.
- At the top of the pons the basilar artery divides into the two posterior cerebral arteries. Each posterior cerebral artery curves around the midbrain to supply the occipital lobe, including the visual cortex, and the inferior surface of the temporal lobe.
- A posterior communicating artery on each side joins the internal carotid artery to the posterior cerebral artery. These two vessels link the anterior and posterior circulations.

Cerebellar and spinal branches

- Three pairs of cerebellar arteries come from the posterior circulation. The posterior inferior cerebellar artery (PICA) is the largest branch of the vertebral artery and supplies the underside of the cerebellum and the lateral medulla. The anterior inferior cerebellar artery (AICA) branches from the lower basilar artery and supplies the front part of the underside of the cerebellum. The superior cerebellar artery branches from the top of the basilar artery, just below the posterior cerebral artery, and supplies the upper surface of the cerebellum.
- The vertebral arteries also give rise to the spinal arteries. A small branch from each vertebral artery joins its partner in the midline to form the single anterior spinal artery, which runs down the front of the spinal cord. The paired posterior spinal arteries arise from the vertebral arteries or from the posterior inferior cerebellar arteries and run down the back of the cord.

Tracing the ring

The ring itself is made of the anterior communicating artery, the first part of each anterior cerebral artery, the

end of each internal carotid artery, the two posterior communicating arteries, and the first part of each posterior cerebral artery. The middle cerebral arteries branch off the ring and are not part of it. Trace the route from a vertebral artery into the ring and around to the front:

1. Vertebral artery: enters the skull through the foramen magnum and runs up the front of the medulla; gives off the posterior inferior cerebellar artery.
2. Basilar artery: formed where the two vertebral arteries unite; runs up the front of the pons, giving off the anterior inferior cerebellar arteries, pontine arteries, and superior cerebellar arteries.
3. Posterior cerebral artery: one of the two end branches of the basilar artery at the top of the pons.
4. Posterior communicating artery: runs forward from the posterior cerebral artery to the internal carotid artery.
5. Internal carotid artery: reaches the base of the brain beside the optic chiasm; gives off the middle cerebral artery laterally.
6. Anterior cerebral artery: runs forward and medially toward the longitudinal fissure.
7. Anterior communicating artery: crosses the midline to the anterior cerebral artery of the other side, closing the front of the ring.

Artery	Comes from	What it supplies
Internal carotid artery	Common carotid artery in the neck	Main source of the anterior circulation; ends by dividing into the anterior and middle cerebral arteries
Anterior cerebral artery	Internal carotid artery	Medial surface of the frontal and parietal lobes, including the leg and foot areas of the motor and somatosensory cortex
Anterior communicating artery	Joins the two anterior cerebral arteries	Closes the front of the ring
Middle cerebral artery	Internal carotid artery (largest branch)	Most of the lateral surface, including the face and upper limb areas of the motor and somatosensory cortex, the auditory cortex, and the insula
Posterior communicating artery	Joins the internal carotid and posterior cerebral arteries	Links the anterior and posterior circulations
Posterior cerebral artery	Basilar artery (end branch)	Occipital lobe, including the visual cortex, and the inferior surface of the temporal lobe
Vertebral artery	Subclavian artery	Main source of the posterior circulation; the two unite to form the basilar artery
Basilar artery	Union of the two vertebral arteries	Brainstem, through the pontine arteries; gives off the anterior inferior and superior cerebellar arteries
Pontine arteries	Basilar artery	The pons
Posterior inferior cerebellar artery (PICA)	Vertebral artery	Underside of the cerebellum and the lateral medulla
Anterior inferior cerebellar artery (AICA)	Basilar artery, lower part	Front part of the underside of the cerebellum
Superior cerebellar artery	Basilar artery, top	Upper surface of the cerebellum
Anterior spinal artery	A branch from each vertebral artery, joined in the midline	Front of the spinal cord
Posterior spinal arteries	Vertebral artery or posterior inferior cerebellar artery	Back of the spinal cord

- The anterior communicating artery joins the two anterior cerebral arteries to each other. The posterior communicating artery is the one that links the front and back circulations.
- The posterior inferior cerebellar artery comes off the vertebral artery. The other two cerebellar arteries come off the basilar. Check where your three start.
- Memory tip: The cerebellar arteries are named from the bottom up: PICA from the vertebral artery, AICA from the lower basilar artery, and the superior cerebellar artery from the top of the basilar artery.

What this means for patients

- Location names the deficit. Because each lobe and cortical area has its own job, the location of an injury predicts the loss. Damage to the occipital lobe affects vision. Damage to Broca's area makes speech slow and effortful, while understanding of language is largely spared.
- The artery involved predicts the pattern too. A stroke in the middle cerebral artery affects the face and arm on the opposite side more than the leg. A stroke in the anterior cerebral artery affects the opposite leg most. A clinician can often estimate where a stroke struck from the pattern of symptoms alone.
- The projection fibers of the internal capsule carry most of the traffic between the cortex and the body through one narrow band. A small stroke there can cut a great deal of wiring at once, which is why such a small lesion can weaken an entire side of the body.
- The Circle of Willis gives the brain a backup route. If one artery feeding the ring narrows slowly, such as one internal carotid artery, blood can sometimes reach its territory through the communicating arteries. This only helps for blockages in the vessels that feed the ring. A blockage in a cerebral artery beyond the ring has no backup route through the circle, and the tissue that artery feeds is at risk. In many people one of the communicating arteries is small or missing, which also limits the backup.

Worked questions

Ten questions in the style of the lecture exam, one for each competency. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 151

On a coronal slice of the cerebrum, the thin darker rim that traces the outline of every fold is the

A. cerebral white matter

B. basal nuclei

C. cerebral cortex

D. corpus callosum

Answer: C. The cerebral cortex is the thin outer layer of gray matter, about 2 to 4 mm thick, and it follows every gyrus and sulcus. Gray matter looks darker because it holds cell bodies, dendrites, and unmyelinated fibers rather than fatty myelin.

Why the others are wrong

- **A** White matter is the pale tissue deep to the cortex, made of bundles of myelinated axons. It is the core, not the rim.
- **B** The basal nuclei are gray matter, but they are islands buried deep inside the white matter, not a rim on the surface.
- **D** The corpus callosum is a thick band of white matter at the bottom of the longitudinal fissure. It is pale, not dark, and it does not follow the folds.

QUESTION 2 · COMPETENCY 152

Which secondary brain vesicle becomes the pons and the cerebellum?

A. Metencephalon

B. Myelencephalon

C. Mesencephalon

D. Diencephalon

Answer: A. The metencephalon, one of the two vesicles from the rhombencephalon, becomes the pons and the cerebellum, and its cavity becomes the upper part of the fourth ventricle.

Why the others are wrong

- **B** The myelencephalon, the other rhombencephalon vesicle, becomes the medulla oblongata and the lower part of the fourth ventricle.
- **C** The mesencephalon does not split. It becomes the midbrain, with the cerebral aqueduct as its cavity.
- **D** The diencephalon comes from the prosencephalon and becomes the thalamus, hypothalamus, and epithalamus around the third ventricle.

QUESTION 3 · COMPETENCY 153

Which groove separates the temporal lobe from the frontal and parietal lobes above it?

A. Central sulcus

B. Parieto-occipital sulcus

C. Longitudinal fissure

D. Lateral sulcus

Answer: D. The lateral sulcus (Sylvian fissure) is the deep groove on the side of each hemisphere. The temporal lobe is below it and the frontal and parietal lobes are above it. The insula is hidden deep inside it.

Why the others are wrong

- **A** The central sulcus runs from the top of the hemisphere down the side and separates the frontal lobe in front from the parietal lobe behind.
- **B** The parieto-occipital sulcus is on the medial surface near the back and separates the parietal lobe from the occipital lobe.
- **C** The longitudinal fissure is the midline cleft between the right and left hemispheres. It does not touch the temporal lobe.

QUESTION 4 · COMPETENCY 154

A patient can understand speech normally but produces slow, effortful speech. Which area is most likely damaged?

A. Wernicke's area, in the temporal lobe

B. Broca's area, in the inferior frontal gyrus

C. Primary auditory cortex, on the superior temporal gyrus

D. Prefrontal cortex, at the front of the frontal lobe

Answer: B. Broca's area plans the movements of speech, so it is a motor area. Damage there makes speech slow and effortful while understanding of language is largely spared, because Wernicke's area is intact.

Why the others are wrong

- **A** Wernicke's area is needed to understand spoken and written language. Damage there would hurt understanding, which this patient has kept.
- **C** The primary auditory cortex receives sound input. Damage there affects hearing, not the production of speech.
- **D** The prefrontal cortex is an association area for planning, judgment, and personality, not for the movements of speech.

QUESTION 5 · COMPETENCY 155

The arcuate fasciculus links Broca's area to Wernicke's area in the same hemisphere. Which class of fiber is it?

A. Association fiber

B. Commissural fiber

C. Projection fiber

D. Corona radiata

Answer: A. Association fibers connect gyri and lobes within the same hemisphere and never cross the midline. The arcuate fasciculus stays inside one hemisphere, so it is an association tract.

Why the others are wrong

- **B** Commissural fibers cross the midline to connect matching areas of the two hemispheres. The corpus callosum is the largest example.
- **C** Projection fibers run up and down between the cortex and the diencephalon, brainstem, and spinal cord. The internal capsule is the example to know.
- **D** The corona radiata is the fan of projection fibers just below the cortex, not a class of fiber and not a cortex-to-cortex link.

QUESTION 6 · COMPETENCY 156

Which nucleus works in the basal nuclei circuits but sits in the midbrain, and is lost in Parkinson disease?

A. Caudate nucleus

B. Globus pallidus

C. Subthalamic nucleus

D. Substantia nigra

Answer: D. The substantia nigra is a band of dark, pigmented neurons in the midbrain, just behind the crus cerebri. It is grouped with the basal nuclei by function, and loss of its neurons is the change seen in Parkinson disease.

Why the others are wrong

- **A** The caudate nucleus is a core basal nucleus in the cerebrum, curving along the lateral ventricle.
- **B** The globus pallidus is the paler, medial part of the lentiform nucleus in the cerebrum.
- **C** The subthalamic nucleus is grouped with the basal nuclei by function, but it sits in the diencephalon below the thalamus, not in the midbrain, and it is not the Parkinson disease nucleus.

QUESTION 7 · COMPETENCY 157

The massa intermedia is a bridge of gray matter that crosses the third ventricle to join the two

A. mammillary bodies

B. hypothalami

C. thalami

D. pineal glands

Answer: C. The thalamus is paired, one egg-shaped mass on each side of the third ventricle. The massa intermedia (interthalamic adhesion) joins the two thalami across the ventricle and shows up as a small cut oval on a midsagittal section.

Why the others are wrong

- **A** The mammillary bodies are two small bumps on the inferior surface of the hypothalamus, behind the infundibulum. Nothing bridges the ventricle between them.
- **B** The hypothalamus forms the floor and lower walls of the third ventricle. The massa intermedia belongs to the thalamus, not the hypothalamus.
- **D** There is one pineal gland, in the epithalamus, projecting back from the roof of the third ventricle.

QUESTION 8 · COMPETENCY 158

Which cerebellar peduncle connects the cerebellum to the pons?

A. Superior cerebellar peduncle

B. Middle cerebellar peduncle

C. Inferior cerebellar peduncle

D. Flocculonodular peduncle

Answer: B. The middle cerebellar peduncle joins the pons. It is the largest peduncle and carries input from the pontine nuclei, which relay signals from the cerebral cortex into the cerebellum.

Why the others are wrong

- **A** The superior cerebellar peduncle joins the midbrain and carries mostly output from the deep cerebellar nuclei toward the midbrain and thalamus.
- **C** The inferior cerebellar peduncle joins the medulla oblongata and carries input from the spinal cord, vestibular nuclei, and olive.
- **D** The flocculonodular lobe is a small lobe on the underside of the cerebellum. There is no peduncle by that name.

QUESTION 9 · COMPETENCY 159

Which limbic structure arches just above the corpus callosum on the medial surface of the hemisphere?

A. Hippocampus

B. Amygdala

C. Parahippocampal gyrus

D. Cingulate gyrus

Answer: D. The cingulate gyrus is the first gyrus above the corpus callosum on a midsagittal section, between the sulcus of the corpus callosum and the cingulate sulcus. It is the part of the limbic ring that sits above the diencephalon.

Why the others are wrong

- **A** The hippocampus is a curved ridge of cortex in the floor of the inferior horn of the lateral ventricle, in the medial temporal lobe, below the diencephalon.
- **B** The amygdala is an almond-shaped nucleus in the front of the medial temporal lobe, deep to the uncus.
- **C** The parahippocampal gyrus is the continuation of the cingulate gyrus down into the temporal lobe, on its inferior and medial surface, not above the corpus callosum.

QUESTION 10 · COMPETENCY 160

The two vertebral arteries unite at the junction of the medulla and pons to form the

A. basilar artery

B. posterior cerebral artery

C. posterior communicating artery

D. internal carotid artery

Answer: A. The basilar artery is formed where the two vertebral arteries unite. It runs up the front of the pons in the midline, gives off the pontine, anterior inferior cerebellar, and superior cerebellar arteries, and splits at the top of the pons into the two posterior cerebral arteries.

Why the others are wrong

- **B** The posterior cerebral arteries are the two end branches of the basilar artery, one step later in the chain.
- **C** The posterior communicating artery is the short vessel on each side that joins the posterior cerebral artery to the internal carotid artery, linking the two circulations.
- **D** The internal carotid artery is the anterior circulation. It comes from the common carotid in the neck and enters the skull through the carotid canal, not by uniting with another artery.